Paleomagnetism of the Mississippi Valley-type ores and host rocks in the northern Arkansas and Tri-State districts

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Received December 12, 1989
Revision accepted February 28, 1990

A paleomagnetic study has been done on 452 specimens from the Early Ordovician Cotter and Middle Ordovician Everton formations in the northern Arkansas Mississippi Valley-type (MVT) lead–zinc ore district and from the Mississippian Warsaw and Keokuk formations in the Tri-State MVT district. Progressive alternating-field and thermal demagnetization first removed a steep normal modern viscous remanence component, leaving an underlying stable remanence of very low intensity (0.3×10^{-5} – 3×10^{-5} A/m). In the Tri-State area, further demagnetization of unaltered host rocks at seven sites in the Warsaw Formation yields a B component with a mean direction of declination (D) 146°, inclination (I) 14° (A_{95} = 13°), corresponding to a paleopole of 130°E, 31°N (Dp = 7°, Dm = 14°), which is a primary Mississippian remanence. In northern Arkansas further demagnetization reveals an A component at 9 sites with a mean direction of D 156°, I 30° (A_{95} = 7°) and a C component at 14 sites of D 160°, I 1° (A_{95} = 9°), corresponding to pole positions of 116°E, 33°N (Dp = 4°, Dm = 8°) and 119°E, 49°N (Dp = 4°, Dm = 9°), respectively. Almost all of the A-component sites are from unmineralized dolomite, whereas all of the C-component sites are from mineralized locations. This correlation suggests that the A magnetization is related to Late Devonian secondary dolomitization and the C magnetization to Permian mineralization produced by brines migrating from the Arkom . basin upon deformation and uplift during the Ouachita orogeny.

Une étude paléomagnétique a été accomplie sur 452 spécimens provenant des formations de Cotter et d'Everton d'âge ordovicien précoce et ordivicien moyen, respectivement, qui incluent les gisements de plomb et de zinc du type de la vallée du Mississippi (TVM) dans le district nord de l'Arkansas, et des formations de Warsaw et de Keokuk d'âge mississippien qui incorporent les gisements de TVM du district Tri-State. La désaimantation progressive par champ alternatif et par chauffage a éliminé en premier une composante subhorizontale d'aimantation rémanente visqueuse, normale et récente, et a laissé une rémanence stable sous-jacente de très faible intensité ($0{,}3 \times 10^{-5} - 3 \times 10^{-5}$ A/m). Dans la région de Tri-State, une autre mesure de désaimantation des roches encaissantes fraîches de sept sites dans la Formation de Warsaw a fourni une composante B avec une direction moyenne de déclination (D) 146°, inclination (I) 14° (A_{95} = 13°), correspondant au paléopôle de 130°E, 31°N (Dp = 7°, Dm = 14°), qui représente une rémanence primaire mississippienne. Dans le nord de l'Arkansas, une étude supplémentaire de désaimantation révèle une composante A pour 9 sites, avec une direction moyenne de D 156°, I 30° (A_{95} = 7°) et une composante C pour 14 sites avec D 160°, I 1° (A_{95} = 9°), correspondant aux positions des pôles de 116°E, 33°N (Dp = 4°, Dm = 8°) et 119°E, 49°N (Dp = 4°, Dm = 9°), respectivement. Presque tous les sites qui ont fourni la composante A apparaissent dans la dolomie nonminéralisée, tandis que tous les sites de composante C sont localisés dans les zones minéralisées. Cette corrélation suggère que l'aimantation A est reliée à la dolomitisation secondaire apparue au Dévonien tardif, et que l'aimantation C est associée à la minéralisation au Permien qui a été produite par la migration de saumures du bassin d'Arkoma poussée par une déformation et un soulèvement durant l'orogenèse d'Ouachita.

[Traduit par la revue]

Can. J. Earth Sci. **27**, 923–931 (1990)

Introduction

Mississippi Valley-type (MVT) lead–zinc deposits are generally thought to have been formed from brines that have migrated from a deep sedimentary basin into peripheral shallow-water carbonate rocks because of sedimentary compaction, tectonic deformation, or uplift (Beales and Jackson 1966; Jackson and Beales 1967; Leach *et al.* 1975; Ohle 1980; Cathles and Smith 1983; Garven 1985; Sverjensky 1986; Leach and Rowan 1986). Unfortunately, the stable tectonic environment in which they characteristically occur precludes age dating by normal geologic methods using intrusive or deformation events. Also, their low temperatures of formation, simple mineralogy, isotopically anomalous Pb, and general lack of K-, Rb-, and U-bearing minerals preclude conventional radiometric age-dating methods. The resulting ambiguity in their age of ore emplacement has prevented significant advances in the understanding of their genesis (Sangster 1983, 1986).

A few paleomagnetic studies have attempted to date MVT mineralization directly or indirectly, but this is the first such study on the northern Arkansas and Tri-State MVT deposits of the American midcontinent (Fig. 1). Beales *et al.* (1974) and Wu and Beales (1981) studied MVT deposits hosted by Upper Cambrian strata in the nearby Viburnum Trend of southeast Missouri. The pole positions they obtained from both ore samples and adjacent host rocks indicated a Late Pennsylvanian – Early Permian (Kiaman) magnetization event, whereas the host rocks remote from mineralization gave poles that correlated better with a Cambrian age. Thus they claimed that the mineralization was Kiaman in age. Subsequently Wisniowiecki *et al.* (1983) found Kiaman poles for both the ores and remote host rocks in the Viburnum Trend, leading them to argue that the mineralization postdated the magnetization event. Also, Dunn and Elmore (1985) isolated a primary Late Cambrian remanence in the Taum Sauk member of the Bonneterre Formation host rocks away from known mineralization, as well as a secondary Kiaman magnetization in dolomitized burrows. Recently Bachtadse *et al.* (1987) studied the east Tennessee MVT district. Their results suggest that the magnetization is actually post-ore in age. Also, many Paleozoic carbonates on the craton have been shown recently to be chemically remagnetized during

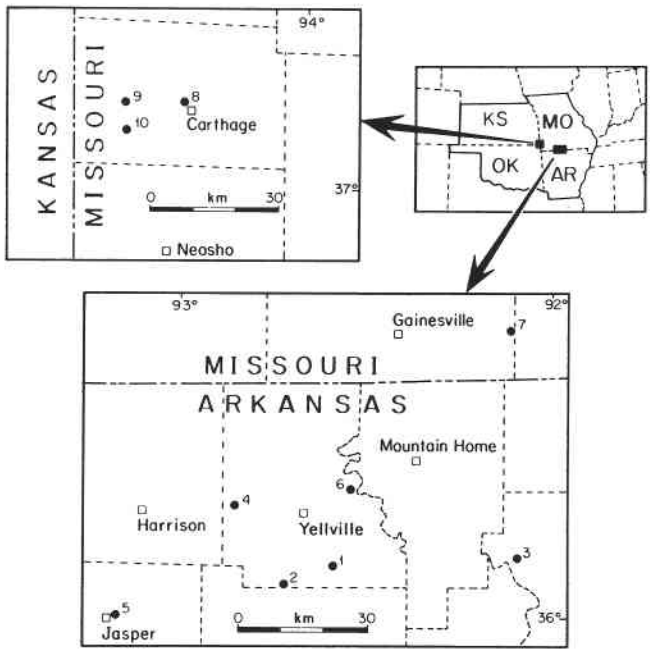


FIG. 1. Sampling locations in the northern Arkansas and Tri-State MVT districts. 1, Mattie May mine; 2, Philadelphia mine; 3, Lucky Dog mine; 4–6, in roadcuts; 7, Alice mine; 8, Carthage limestone quarry; 9, Oronogo Circle mine; 10, King Jack Park mine.

the late Paleozoic by the diagenetic development of secondary magnetite (Scotese *et al.* 1982; McCabe *et al.* 1983; Bachtadse *et al.* 1987). The geologic setting and mineralization characteristics of the northern Arkansas and Tri-State MVT deposits differ somewhat from those of southeast Missouri and east Tennessee. By comparing paleomagnetic results from them, it may be possible to better date the age of the MVT mineralization and thus further constrain possible hypotheses for their genesis. For example, if all of these districts give Kiaman poles for both the ore and remote host rocks, then it is probable that tectonism and uplift associated with the Alleghenian and Ouachita orogenies were responsible for decanting deep basin brines, as Leach and Rowan (1986) have proposed.

Geology

The northern Arkansas and Tri-State MVT deposits are located in a structural platform that is bounded on the north by the Ozark uplift and on the south by the Arkoma basin and Ouachita uplift. Their sedimentary rocks are mostly Cambro-Ordovician carbonates in northern Arkansas and Mississippian carbonates in Tri-State. In northern Arkansas the deposits are largely found in Ordovician limestones of the Cotter and Everton formations (Suhm 1974). In contrast, the Tri-State deposits are found almost entirely in limestones of the Mississippian Keokuk and Warsaw formations. In both districts the primary ore minerals include sphalerite, galena, dolomite, quartz, pyrite, and jasperoid, with sphalerite being more abundant than galena. Presumably these minerals were preferentially precipitated in the most permeable, coarsely crystalline carbonate rocks in the sequences. Most of the ore occurs as open-space fillings in breccias and as replacement zones adjacent to faults and breccia zones (McKnight 1935; Brockie *et al.* 1968). All of the faults in northern Arkansas postdate early Pennsylvanian, whereas those in Tri-State are both pre-ore and post-ore. The Mississippian limestones in the Tri-State are overlain unconformably by the Pennsylvanian Cherokee shale, which contains not only the youngest strata in the district but also small MVT deposits in limestone interbeds. These deposits in Tri-State and those in Pennsylvanian faults in northern Arkansas indicate that at least some mineralization must be Pennsylvanian or younger.

Sphalerite fluid-inclusion studies indicate that freezing temperatures for northern Arkansas and Tri-State range from −20.8 to −23.6°C and from −23.0 to −23.4°C respectively, corresponding to salinities of greater than 22 wt% salt (Roedder 1967; Leach *et al.* 1975). Their homogenization temperatures range from 83 to 120°C and from 83 to 132°C, respectively (Schmidt 1962; Leach *et al.* 1975). The similarity of the salinities, temperatures, and stratigraphic relationships led Leach *et al.* (1975) and Leach and Rowan (1986) to suggest that the northern Arkansas and Tri-State MVT deposits were genetically related and represented the same period of mineralization.

Sampling and experimental techniques

Four block samples were collected from most sites after orienting with a Brunton compass. In northern Arkansas, 19 sites were taken in the Everton Formation and 11 sites in the Cotter Formation (Fig. 1; Table 1); for Tri-State, 8 sites were taken in the Warsaw Formation and 10 sites in the Keokuk Formation (Fig. 1; Table 2). Most of the unaltered host rocks are limestones or diagenetic dolostones, whereas most of the "mineralized" rocks—although of the same lithologies—contain visible sphalerite and (or) galena and (or) jasperoid. The latter is a widespread form of wall-rock alteration considered to be genetically and temporally related to the ore (Brockie *et al.* 1968). Samples were selected from fresh road cuts, mine pits, and quarries to avoid weathering. Even so, some sites seem to have been affected by a modern weathering.

Several core specimens with a 2.5 cm diameter and 2.2 cm length were drilled from each block to give a total of 452 specimens. Measurements were done inside a shielded room with an ambient magnetic field of about 100 nT. The specimens were stored inside this room for several days before measurement to reduce their acquired viscous remanent magnetization (VRM). The natural remanent magnetization (NRM) was measured using an automated two-axis CTF cryogenic magnetometer with a noise level of $<4\text{x}10^{-6}$ A/m. The NRM of the specimen holder was first measured and then subtracted from each specimen measurement to remove its contribution. Because of the very low NRM intensities in the range of $0.1 \times 10^{-4} - 3 \times 10^{-4}$ A/m for northern Arkansas and of $1 \times 10^{-5} - 8 \times 10^{-5}$ A/m for Tri-State, some other measures, such as the routine demagnetization and alcohol cleaning of the specimen holder and the continuous nulling of the sensors, were adopted to minimize possible errors. Alternating field (AF) demagnetization was done using a Sapphire Instruments SI-4 unit and thermal demagnetization was done using a Schonstedt TSD-1 demagnetizer.

One pilot specimen from each block was AF demagnetized in 14 steps to 160 mT and a second specimen was thermally demagnetized in 50°C or 25°C steps until the measured magnetization became erratic or fell below the noise level of the magnetometer. The specimens behaved similarly on detailed AF and thermal demagnetization. Therefore, the remaining specimens were AF demagnetized only, using at least four steps. Demagnetization paths were plotted on orthogonal vector diagrams (Zijderveld 1967) and stereograms. Results were evaluated by visual inspection and then directions were determined by principal-component analysis (Kirschvink 1980) with

TABLE 1. Summary of paleomagnetic data from the northern Arkansas district

Site no.	N	N_0	D (°)	I (°)	MAG	A_{95} (°)	k	Demagnetization range represented
Everton Formation								
Mattie May mine								
1	10	12	178	+73	—	23	5	80 mT, 530°C
2	5	10	142	+3	C	25	10	30–80 mT, 400°C
3	7	9	163	+1	C	50	3	40–80 mT, 530°C
4	5	11	180	−6	C	63	2	60–80 mT, 500°C
5	6	13	174	+34	A	26	8	15–80 mT, 300–450°C
6	8	10	321	+79	—	34	4	20–80 mT, 500°C
7	9	11	136	+8	C	17	11	40–80 mT
Philadelphia mine								
8	6	10	153	−2	C	17	16	60–100 mT, 400°C
9	8	12	151	+1	C	23	7	0–100 mT, 450°C
10	6	11	190	+8	C	46	3	20–140 mT, 575°C
11	3	9	181	−18	C	29	19	80–140 mT, 450°C
Lucky Dog mine								
12	6	10	218	+2	—	26	8	40–80 mT, 400°C
13	8	15	154	−1	C	23	7	40–80 mT, 450°C
14	7	10	148	0	C	14	18	60–80 mT
15	5	10	155	−5	C	17	22	60–80 mT, 450°C
16	9	10	163	+10	C	24	6	60–80 mT, 500°C
Road cuts								
17	10	10	154	+3	C	15	12	60–100 mT, 500°C
Cotter Formation								
New road cuts								
20	10	10	189	+65	—	21	6	60–100 mT, 500°C
21	4	9	161	+15	A	16	36	80–100 mT, 450°C
22	10	10	147	+33	A	8	36	80–160 mT
23	12	12	147	+33	A	9	23	80–160 mT
24	14	14	154	+20	A	11	15	80–140 mT, 500°C
25	8	8	147	+39	A	11	25	80–140 mT, 500°C
Alice mine								
26	7	7	66	+54	—	27	6	80–140 mT, 500°C
27	11	11	163	+27	A	15	10	80–120 mT, 530°C
28	7	8	155	+31	A	21	9	80–120 mT
29	8	8	175	+9	C	10	32	80–120 mT, 500°C
30	11	11	150	+37	A	10	23	80–140 mT, 500°C
A component								
Mean:	D 156°, I +30°, A_{95} = 7°, k = 53, N = 9							
Pole position:	33°N, 116°E, Dp = 4°, Dm = 8°							
C component								
Mean:	D 160°, I +1°, A_{95} = 9°, k = 22, N = 14							
Pole position:	49°N, 119°E, Dp = 4°, Dm = 9°							

NOTES: Sites 18 and 19 did not retain a coherent remanence; N is the number of specimens used to compute the site mean; N_0 is the number of specimens measured; the mean remanence direction is given by its declination (D) and inclination (I); MAG is the isolated magnetization component; A_{95} is the radius of the cone of 95% confidence; k is the precision parameter of Fisher (1953); Dp and Dm are the radii of the ellipse of 95% confidence along and perpendicular to the direction of the magnetic meridian from the site to the pole, respectively.

maximum angular deviation values of less than 15°. Mean directions and poles were calculated using standard Fisher (1953) statistics.

Results

Northern Arkansas

The initial NRM directions of all specimens are scattered, with most being steeply downwards to the southeast (Fig. 2A). No difference was found between the NRM directions of the unaltered host and mineralized rocks, although the former are generally better clustered. During thermal step demagnetization above 500°C, most specimens showed an erratic behaviour, probably because of the conversion of magnetite to hematite and (or) pyrite to magnetite (Lowrie and Heller 1982). Data from specimens that appeared to have been so altered were rejected before identifying those specimens with a characteristic magnetization direction. The specimens from sites 1, 6, 20, and 26 proved to have an univectorial magnetization with a very steep normal inclination (Fig. 3A). Because these steep directions parallel the present Earth's magnetic field (PEMF) direction,

TABLE 2. Summary of paleomagnetic data from the Tri-State district

Site no.	N	N_0	D (°)	I (°)	MAG	A_{95} (°)	k	Demagnetization range represented
				Warsaw Formation				
Carthage quarry								
31	5	7	144	+29	B	11	54	40–80 mT, 350°C
32	6	10	153	+30	B	19	18	20–80 mT
33	7	8	151	−10	B	20	10	40–100 mT, 350°C
34	7	7	131	+12	B	25	7	20–60 mT
35	6	7	151	+29	B	23	9	20–100 mT, 350–500°C
36	6	12	137	+13	B	17	16	60–100 mT, 500°C
38	6	9	131	+30	B	16	19	20–60 mT, 400–500°C
A component								
Mean:	D 142°, I +19°, A_{95} = 13°, k = 13, N = 7							
Pole position:	31°N, 130°E; Dp = 7°, Dm = 14°							

NOTES: Site 37 did not retain a coherent remanence. See Table 1 for explanation of the symbols.

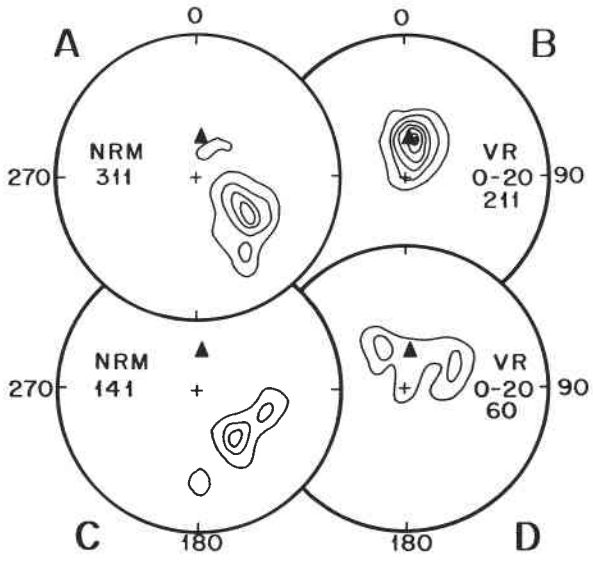


FIG. 2. Contour plots of the percentage density for the measured vectors (number given in stereogram) plotted on the lower hemisphere of equal-area stereograms: (A) NRM directions for northern Arkansas; (B) remanence vectors removed between 0 and 20 mT for northern Arkansas; (C) NRM directions for Tri-State; (D) remanence vectors removed between 0 and 20 mT for Tri-State. The triangle is the PEMF direction. The outer contour is the 95% confidence bound of about 3.6% density and the contours increase in 2% increments from it (Kamb 1959; Stupavsky and Symons 1982).

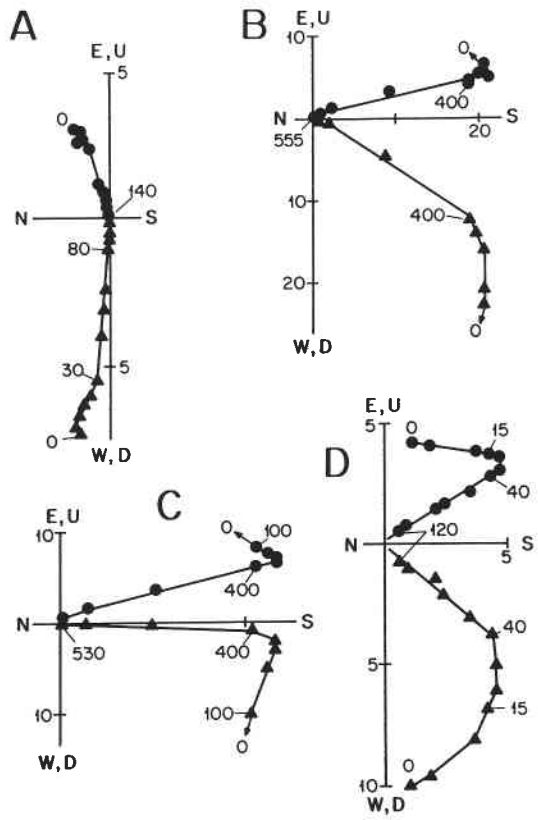


FIG. 3. Orthogonal demagnetization diagrams of example specimens from northern Arkansas showing removal of (A) a vertical, recent PEMF component from site 20 on AF demagnetization; (B) an A component on thermal demagnetization from site 27; (C) a C component on thermal demagnetization from site 29; (D) an A component on AF demagnetization from site 24. Circles and triangles denote projections on the horizontal and vertical planes respectively. The axis unit is in 10^{-5} A/m. The step intensities for AF and thermal demagnetization are in mT and °C, respectively.

these sites were deemed to be severely altered by modern weathering and were rejected from further statistical analysis.

For most of the remaining northern Arkansas specimens, a two-component magnetization could be clearly identified on both AF and thermal step demagnetizations (Figs. 3, 4A, 4B). First a steeply inclined component with relatively low coercivity or low unblocking temperature is usually removed by 20 mT on AF demagnetization or by 350°C on thermal demagnetization. Figure 2B is a contour plot of the density of the vectors removed between 0 and 20 mT. It shows that this low coercivity component is aligned with the PEMF direction, identifying it as a Holocene VRM.

Further demagnetization reveals a higher coercivity or unblocking temperature component that is typically found in the 20–100 mT or 350–500°C range. This underlying magnetization is directed to the southeast and forms two significantly different clusters, one with a moderate inclination, called the A component, and the other with a shallow inclination, called the C component.

The A component was isolated in specimens from nine sites, mostly from the Cotter Formation (Figs. 3B, 3D). Five are from new road cuts in unaltered limestone or dolomite. Although the

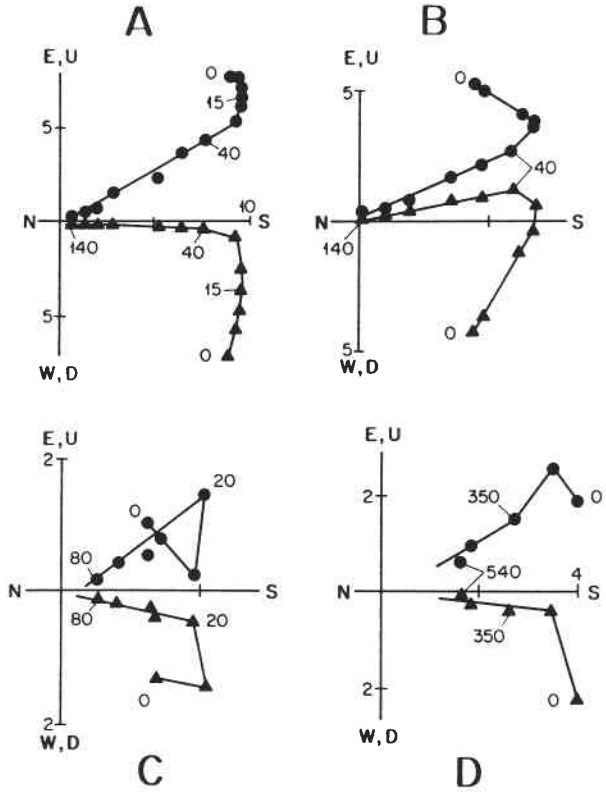


FIG. 4. Orthogonal demagnetization diagrams of example specimens from northern Arkansas and the Warsaw Formation in the Tri-State district showing: (A) a C component on AF demagnetization from site 29; (B) a C component on AF demagnetization from site 8; (C) a B component on AF demagnetization from site 33; (D) a B component on thermal demagnetization from site 35. Conventions as in Fig. 3.

Alice mine samples were collected from pits, the samples were mostly unmineralized, so that the A component was found at three sites and the C component was found at only one site (29). In contrast, the specimens from the mineralized Everton Formation showed a predominance of the C-component magnetization, with 13 of 19 sites retaining it (Figs. 3C, 4A, 4B). Site 5 in unmineralized limestone from the Mattie May mine is the lone exception and exhibits the A component.

McCabe *et al.* (1984) studied the Middle Ordovician Trenton limestone and compared the results with the earlier investigation by McElhinny and Opdyke (1973). McCabe *et al.* (1984) found that the direction obtained from thermal demagnetization was shallower than that from AF demagnetization, indicating that thermal demagnetization was more effective in removing the modern VRM component. For northern Arkansas, we averaged the directions obtained from thermal and AF demagnetization separately for the A component and did not find any significant difference between them, showing that AF demagnetization is as effective as thermal demagnetization in isolating the characteristic direction.

Tri-State

The NRM directions of all samples are widely dispersed except for those from the unaltered Warsaw Formation. The latter form a cluster with a moderate inclination to the southeast, indicating that their remanences are also relatively stable (Fig. 2C). For most specimens the thermal demagnetization results proved unsatisfactory, yielding erratic remanence directions and intensities above 450°C. As before, these features can be

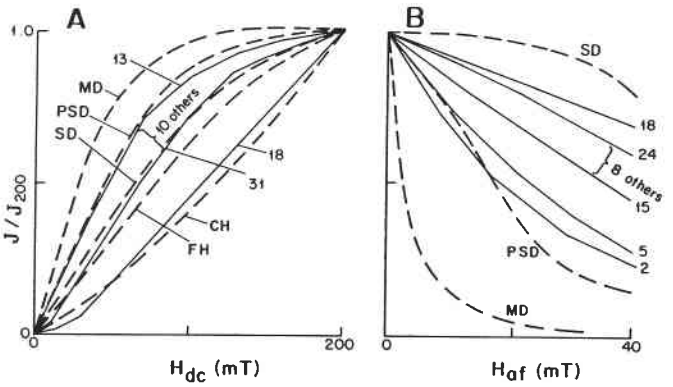


FIG. 5. Isothermal remanent magnetization test results for specimens from the northern Arkansas and Tri-State districts: (A) acquisition of IRM; (B) AF demagnetization of IRM. The type curves are for single domain (SD), pseudosingle domain (PSD), and multidomain (MD) magnetite and for fine-grained hematite (FH) and coarse-grained hematite (CH) (Dunlop 1971, 1972, 1973, 1981).

ascribed to chemical or mineralogical changes of the magnetic carriers on heating. Such changes are evident because these specimens redden and become more intensely magnetized by 600°C. Therefore, all remaining specimens were treated by AF demagnetization.

Although the intensities are very weak, two magnetization components could still be successfully isolated for most of the Warsaw Formation specimens from the Carthage quarry (Figs. 4C, 4D). Typically they have a low-coercivity component that is removed by 10–20 mT on AF demagnetization, which we interpret to be a VRM overprint from the PEMF (Fig. 2B). Further demagnetization above 20 mT usually produces a straight-line segment towards the origin on a vector-component diagram (Figs. 4C, 4D). Where the vectors do not converge to the origin, the increasing scatter is mainly caused by their decreasing intensities approaching the noise level of the magnetometer. For these cases the preceding linear segment was extrapolated to the origin to get the characteristic direction. In seven of the eight sites from the unaltered Warsaw limestone, a characteristic direction, called the B component, was clearly isolated. Site 37 gave incoherent directions and so was excluded (Table 2). The remaining specimens from the Keokuk Formation were mostly well mineralized with some containing unavoidable traces of the abundant modern hematitic weathering within the pits. Although their NRM intensities are slightly stronger than those from the Warsaw limestone, they either contained an unstable magnetization with no directional consistency or gave very steep directions close to the PEMF direction during demagnetization. We did not analyse them further.

Rock magnetism

Isothermal remanent magnetization

For northern Arkansas, 11 representative specimens were selected for isothermal remanent magnetization (IRM) testing. The IRM acquisition curves for both A- and C-component specimens show a rapid, smooth increase in intensity. This behaviour, combined with their behaviours on the thermal and AF demagnetizations, is typical of single-domain (SD) to pseudosingle-domain (PSD) magnetite remanence carriers (Fig. 5A). The specimen from site 18 with an anomalous remanence direction yields an almost linear increase in IRM intensity that is typical of hematite (Fig. 5A) and is probably from modern weathering. Alternating-field demagnetization of the IRM

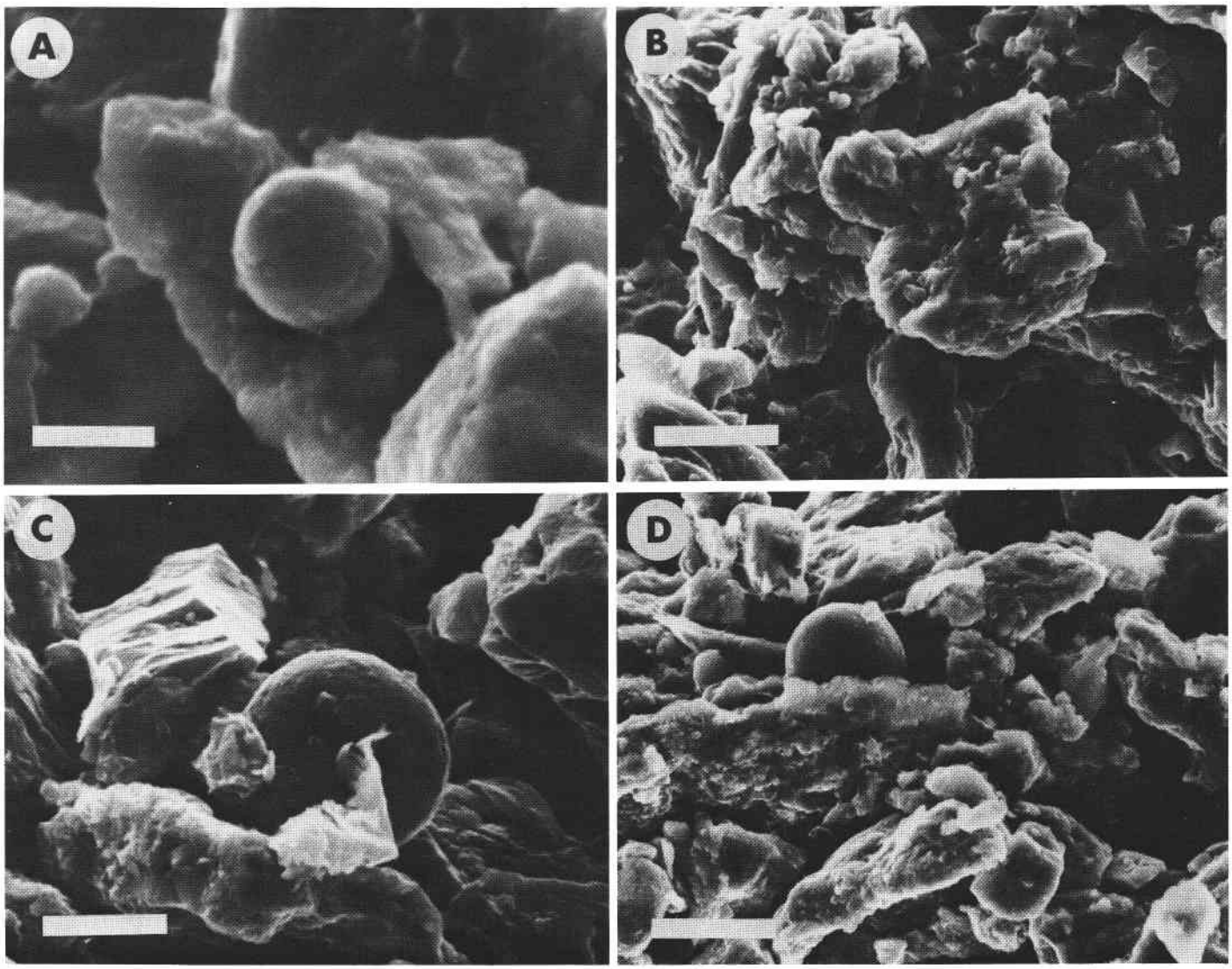


FIG. 6. Scanning electron micrograph of magnetite crystals extracted from: (A) and (B) A-component sites, scale bars are 2 and 10 μm respectively; (C) B-component site, scale bar is 5 μm; (D) C-component site, scale bar is 10 μm.

shows their SD to PSD character even more clearly (Fig. 5B). The relatively high coercivities found on AF demagnetization also indicate SD or PSD magnetite domains.

The IRM acquisition and decay curves for six Tri-State specimens are similar to those of northern Arkansas and indicate that their remanence carriers are dominantly SD magnetite (Fig. 5).

Magnetic extracts

Magnetic mineral extracts were separated from seven samples to examine their morphology and chemical compositions using a scanning electron microscopy (SEM) with energy dispersive analysis (EDA). The extracts represented a variety of rock types and magnetic components. Each was prepared by crushing, grinding, and dissolving four core specimens in buffered acetic acid of pH = 4.5 for several days. The extracts were obtained by pumping the dilute water suspension of insoluble residues slowly through an electromagnetic separator. The extracts were then washed with distilled water into mounting stubs for the SEM and EDA studies.

Scanning electron microscopy revealed the dominant presence of angular blocky crystals in all seven extracts, which, based on EDA, show iron as the main element with minor silicon (Fig. 6A). No titanium, nickel, or sulphur was found by EDA, indicating that these blocky crystals are likely quite pure magnetite. Perfect spheroids with iron as the only detectable element were also found in five of the extracts (Figs. 6B, 6C, 6D). They are likely magnetite but also could be wustite. These spheroids are from 1 to 6 μm in diameter. The extracts from the C-component sites contain several spheroids with a range of sizes, whereas the A- and B-component extracts had only one or two small spheroids each. These spheroids are similar to those reported by McCabe *et al.* (1983, 1987) and Elmore *et al.* (1987), but are much smaller in size and quantity. They considered the spheroids to be diagenetic in origin. However, the presence of these spheroids in both mineralized and unaltered host rocks with the A, B, and C components suggests that their genesis may be related to some other mechanism. Although most spheroids are big enough to be multidomain carriers (Dunlop 1981), their surfaces are subdivided by crystalline patterns of PSD or SD size, as McCabe *et al.* (1983) found. Thus they may be stable magnetic carriers as the IRM tests show. Our observation that the spheroids are more abundant in mineralized than in unmineralized sites suggests

that at least some spheroidal magnetite was formed during the mineralization event. Nevertheless, blocky magnetite crystals appear to be the only significant remanence carriers in the unmineralized sites and the most significant carriers in the mineralized sites.

Discussion

This study has revealed two significantly different magnetic-component directions for northern Arkansas. The mean of the nine site means with the A component is declination (D) 156°, inclination (I) 130° (A_{95} = 7°) (Fig. 7A); and the mean of the 14 site means with the C component is D 160°, I 1° (A_{95} = 9°) (Fig. 7C). The A and C components are both reversed in polarity. When inverted, the A component yields a pole position at 116°E, 33°N (Dp = 4°, Dm = 8°) and the C component, at 119°E, 49°N (Dp = 4°, Dm = 9°). The A and C poles, when compared to the apparent polar wander path (APWP) for cratonic North America (Van der Voo 1990; Fig. 8), suggest a Late Devonian age and a Permian age, respectively. Sampling limitations in these flat-lying rocks precluded field tests to further constrain the magnetization ages. Thus we can only infer relationships between the magnetizations and mineralization from geological arguments and magnetization characteristics.

Almost all of the A-component sites are from unmineralized dolomite sites, whereas all of the C-component sites are from mineralized locations. This correlation indicates that the A magnetization is related to secondary dolomitization in the Late Devonian and the C magnetization to the mineralization event in the Permian. This is consistent with geological evidence that indicates that the primary micrite carbonates were extensively dolomitized before mineralization, principally by the incomplete cementation of primary porosity and to a lesser extent by replacement of the micrite (Long *et al.* 1986); this dolomitization event could have produced a chemical remanent magnetization (CRM). Also, the presence of authigenic magnetite spheroids in the dolomites is not necessarily inconsistent with this hypothesis. Hart and Fuller (1988) reported similar authigenic magnetites in the Monterey dolomites of California and considered that they were formed during late-stage diagenesis, as might authigenic hematite in some cases (Elmore *et al.* 1985). However, it is possible that the formation of authigenic magnetite in a carbonate may also occur during dolomitization.

Although we cannot prove conclusively that the mineralization itself carries the C component, the geological evidence and the magnetization association indicate that these two events are coeval. The widespread occurrence of trace MVT mineralization, combined with the fluid inclusion and geochemical data, suggest that the mineralization was emplaced on a regional scale throughout the Ozarks (Leach 1979; Coveney *et al.* 1987). Specifically, the MVT deposits were related to the migration of brine fluids out of the deep Arkoma foreland basin and into the midcontinental shallow marine carbonates when the foreland basin was deformed and uplifted to create the Ouachita belt during the Pennsylvanian–Permian collision of the Llanorian and North American plates in the Ouachita orogeny (Leach and Rowan 1986; Oliver 1986). The late Paleozoic was a time of widespread tectonic activity, as both the Ouachita and Alleghenian orogenies caused intraformational brines to migrate through the Palaozoic strata of the midcontinent. This migration is recorded in many paleomagnetic studies as a Kiaman pole through the complete remagnetization of both unmineralized and mineralized carbonates (Scott 1979; McCabe *et al.* 1983; Wisniowiecki *et al.* 1983; Bachtadse *et al.* 1987; this study). Long *et al.* (1986) studied the relationship between solution collapse and mineralization in northern Arkansas. They found that the early stage of brecciation was formed by groundwater dissolution after Early Mississippian time and that the brecciation clearly preceded mineralization. Hayes *et al.* (1989), using ultraviolet fluorescence techniques, have shown that the hydrothermal dolomites associated with ore in northern Arkansas and the Viburnum Trend – Old Lead Belt of southeastern Missouri are the same and differ from "non-ore" hydrothermal dolomites, further indicating that both districts were mineralized at the same time. The occurrence of sphalerite and galena in Pennsylvanian shales and faults demands that at least some of the mineralization must be Pennsylvanian or younger. Although a multiple-event mode of mineralization is possible, it is equally or more likely that only one event was responsible for the formation of the MVT deposits. This is supported by the paleomagnetic data to date, which show only Kiaman magnetizations for MVT mineralized material. Therefore we conclude that the MVT mineralization event coincides with the acquisition of the C component during the Late Pennsylvanian – Permian in the Kiaman reversed superchron.

The simplest explanation for the origin of the C component is

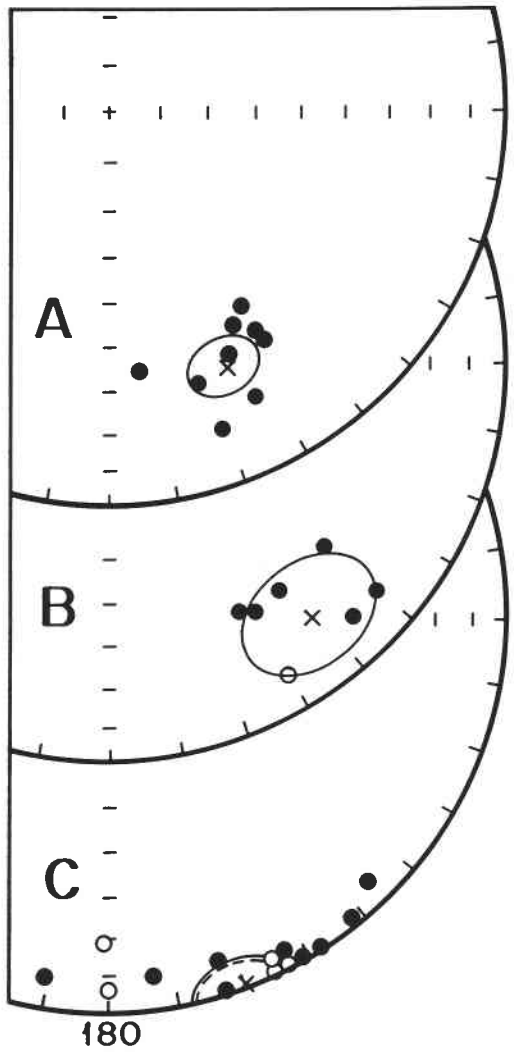


FIG. 7. Equal-area stereograms for the northern Arkansas and Tri-State districts showing site mean directions after demagnetization for the: (A) A component; (B) B component; (C) C component. The component mean direction (X) is circumscribed by its cone of 95% confidence. Open symbols are in the upper hemisphere and closed in the lower. See also Tables 1 and 2.

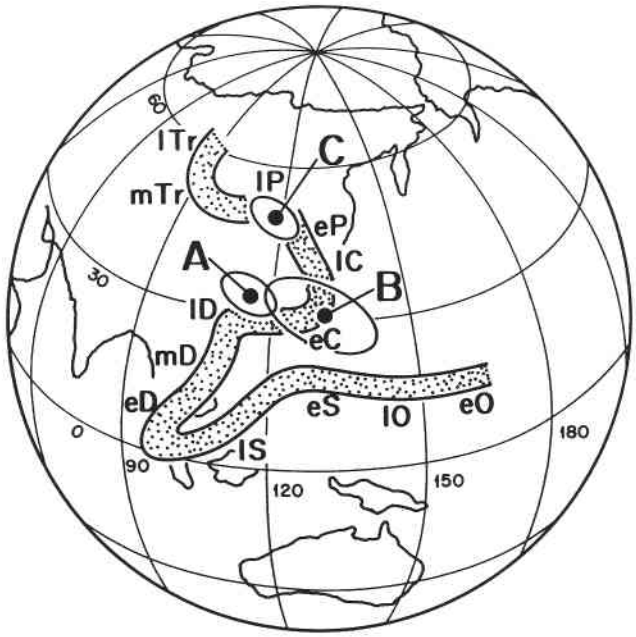


FIG. 8. Pole positions for the A, B, and C components with their ovals of 95% confidence. The apparent polar wander path is from Van der Voo (1990) and starts in Early Ordovician (eO) and ends in Late Permian (lP).

that it is a CRM related to the mineralization event. In northern Arkansas, temperatures for fluid inclusions in sphalerite, quartz, and dolomite range predominantly between 95 and 135°C (Leach *et al.* 1975). The fact that these inclusions have not been stretched or leaked implies that these temperatures have not been exceeded since the mineralization event. The preservation of well-formed marcasite crystals on sphalerite and dolomite crystals also indicates that the mineralized rock was not subsequently heated above 250°C, otherwise the marcasite would have inverted to pyrite (Skinner 1967). These temperatures are too low to have thermally remagnetized preexisting magnetite with unblocking temperatures above 540°C (Walton 1980). Our finding that authigenic spheroidal magnetite seems to be more abundant in mineralized specimens than in unaltered host rocks and secondary dolomite specimens suggests that most, if not all, of the spheroids were formed during a secondary CRM event. The genesis of the authigenic magnetite carrying the C component is still unknown. Recent studies by Elmore *et al.* (1987) and McCabe *et al.* (1987) found authigenic spheroidal magnetite in solid bitumen and in calcite speleothem deposits. Thus they suggest a genetic relationship between hydrocarbon migration and the precipitation of authigenic magnetite, probably as a result of microbial attack on the hydrocarbons. In northern Arkansas and Tri-State, the jasperoids appear to have been deposited contemporaneously with the MVT ore. Most of these jasperoids contain brown or black opaques of bitumen or other organic matter (Smith and Siebenthal 1907; Long *et al.* 1986). Thus the migration of hydrocarbons may be related to the precipitation of the authigenic magnetite during the MVT mineralization event, as proposed by Elmore *et al.* (1987) and McCabe *et al.* (1987). Further, Oliver (1986) cites geologic evidence that tectonic uplift during the late Paleozoic was also accompanied by the abundant migration in the midcontinent of petroleum with the hot brines.

For the Warsaw Formation in the Tri-State district, a mean direction of D 142°, I 19° (A_{95} = 13°) has been obtained from the seven sites with the B component (Fig. 7B). The corresponding pole position is at 31°N, 130°E (Dp = 7°, Dm = 14°). This B component appears to be primary, although there are no field tests to rigorously constrain its age. However, there are several reasons to suggest that it was probably acquired during or immediately after deposition of the limestone. First, the B paleopole is indistinguishable from the Mississippian segment of the APWP for North America (Fig. 8), which agrees with the age of the Warsaw Formation. Second, the limestone from the Carthage quarry is unaltered and several kilometres away from known mineralization. Third, neither traces of fluid nor recrystallization have been detected in these very "tight" limestones (T. S. Hayes, personal communication, 1988). Thus it is very likely that they have escaped pervasive remagnetization by hot basinal brines to preserve a primary magnetization. In Tri-State the temperatures of sphalerite fluid inclusions range from 83 to 120°C (Schmidt 1962). For the same reasons as cited for northern Arkansas, a viscous thermal remagnetization origin can be ruled out for the B remanence component. The trace occurrence of authigenic magnetite in the limestone indicates that the B component may be a chemical remanence formed during lithification or induration of the limestone (Lowrie and Heller 1982).

Acknowledgments

The authors wish to thank D. L. Leach, T. S. Hayes, and J. G. Viets of the United States Geological Survey and R. M. Coveney, Jr., of the University of Missouri at Kansas City for their great help in the field collecting for this study. We also thank A. D. Chiasson, who assisted in the field and prepared the specimens for measurement, and J. W. Robinson of the University of Windsor, who helped with the SEM–EDR analysis. The manuscript benefitted from the many useful comments of D. L. Leach, J. Hodych and R. Van der Voo. This study was funded through a research agreement with the Department of Energy, Mines and Resources Canada.

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